

SCHOOL OF COMPUTING (SOC)

Project2 Specification

DIPLOMA IN INFORMATION TECHNOLOGY

ST0528 Database Systems (Project)

2026/2027 Semester 1

Instructions and Guidelines

1. This is a **GROUP** assignment.
2. The deadline of this assignment is on **14 Aug 2026 Friday (6:30 pm)**.
3. You are REQUIRED to do the following for your submission:
 - Deliverable #001 : Code Commits on Github Classroom repository. It will be reviewed for marking.
 - Deliverable #002 : Completed Project2 Individual Report. The implementation will be performed as a group but **the report will individual**.
 - Deliverable #003 : Zip file of your source code.
 - Deliverable #004 : Zip file of a backup of the final state of your database.
 - Declaration form : A word document for declaration of academic Integrity

The final submission on BrightSpace is a single zip file of deliverables #002, #003, #004 and the declaration form with the following naming convention:

YourStudentID-YourName.zip

2012345-DavidTan.zip

4. Submission requirements for **deliverable #001: Commits on Github Classroom repository**
 - All code submissions must be made through GitHub Classroom.
 - You are advised to practice version control by committing your code regularly as you make progress. Regular and clear commits demonstrate a well-organized development process, help track changes and make it easier to revert to previous states if needed.
 - **Maintain a stored_procedures.sql file** for the database-level stored procedures and/or functions that you had used. Make sure to **commit this file** as part of your final commit before submission.
 - Submissions that do not adhere to the version control and submission guidelines would be subjected to a **penalty**.
5. Submission requirements for **deliverable #002: Project2 Individual Report**
 - The template is available in BrightSpace under Assignments > Project2 > Project2 Individual Report.
 - The report should include clear documentation of evidence to support work done for each section listed in the template.
 - It should be submitted as a PDF document using the following file-naming convention

YourName_Project2Report.pdf

DavidTan_Project2Report.pdf

6. Submission requirements for **deliverable #003: Source codes of modified application**
 - Should be submitted as a zip document using the following file-naming convention
YourName_Code.zip
DavidTan_Code.zip
 - The zip file should contain source codes of your application (**without the node_modules folders**)
7. Submission requirements for **deliverable #004: Backup of database**
 - Should be submitted as a zip document using the following file-naming convention
YourName_Db.zip
DavidTan_Db.zip
 - The zip file should contain the backup SQL file of your database. The instructions on how to backup can be found under “How to backup a database” on BrightSpace > Resources > Guides
8. **Warning: No marks** will be awarded, if the work is copied or you have allowed others to copy your work. **Plagiarism** means passing off as one's own the ideas, works, writings, etc., which belong to another person. In accordance with this definition, you are committing plagiarism if you copy the work of another person and turning it in as your own, even if you would have the permission of that person. Plagiarism is a serious offence, and if you are found to have committed, aided, and/or abetted the offence of plagiarism, disciplinary action will be taken against you. If you are guilty of plagiarism, you may **fail** all modules in the semester, or even be liable for **expulsion**.
9. **Late penalty: 50%** of the marks will be deducted for assignments that are received within **ONE (1)** calendar day after the submission deadline. No marks will be given thereafter. Exceptions to this policy will be given to students with **valid LOA on medical or compassionate grounds**. Students in such cases will need to inform the lecturer as soon as reasonably possible. Students are **not to assume** on their own that their deadline has been extended.
10. A presentation/interview will be conducted after the code submission. **No marks** will be awarded if you are absent for the interview.

Overview

This project focuses on the implementation of a web application to digitize and streamline the vision-screening workflow across multiple stations, enabling real-time participant tracking, efficient data entry and authorized access controls.

A. Requirements

Baseline Requirements:

Refer to the Common Project Specification document.

Project Management

This refers to the collaboration and means to manage the project. It includes planning, task allocation, collaborative decision-making, tracking of progress and use of version control to maintain steady progress throughout the development.

Database Design and Data Modeling

- Design and model your entities via Prisma ORM models
- Define and clearly document all entities, attributes, constraints and relationships among the entities
- ERD(s) should be diagrammed on Lucid Chart and iterations should be included as clear images in the report. Include a **link to the final version of your ERD on Lucid Chart**.
- Provide explanations of your design choice(s)

Database Operations Implementation

Implement the necessary database operations using the Prisma Client, database stored procedures and functions. Your implementation should include the creation of the relevant database tables with appropriate tables, data types and constraints using consistent naming conventions.

Application Development

Integrate the PostgreSQL database with a web application. The web application should be able to interface with data in the database and effectively support the baseline requirements stipulated.

B. Evaluation

Project Management 15%	Evaluates the systematic planning, organization, and execution of the project. This includes the means of managing the project including the planning, task allocation, tracking of progress, collaborative-decision making and use of version control to maintain steady progress throughout the development. Key Aspects • Clear definition of scope and task distribution • Consistent work and use of version control • Evidence of iterative progress, task tracking • Constructive participation and inclusive decision making
Database Design & Data Modeling 25%	Evaluates the soundness and completeness of the database schema design and its representation using Prisma ORM. This includes the definition of models, relationships, fields, types, and constraints to ensure data integrity and support application requirements. Key Aspects • Logical and complete representation of the database schema • Proper normalization and appropriate use of relations, constraints, indexes, and data types to maintain data integrity and optimize performance. • Alignment of the data model with the requirements of the application
ORM and Stored Procedures Implementation 15%	Evaluates the implementation of database operations using Prisma Client alongside database-level stored procedures and functions. This includes writing efficient queries and transactions to handle complex data manipulation and business logic, ensuring a secure and performant data layer. Key Aspects • Correct and efficient use of Prisma Client for queries • Well-structured stored procedures/functions for database-level operations where appropriate • Robust error handling of database queries that efficiently support features and their complexities.
Application Development 20%	Evaluates the functionalities of the web application and the quality of the code implementation. This includes the development of the relevant features based on clean, well-organized code that adheres to coding best practices. The application should be user-friendly and reliable. Key Aspects • Implementation of relevant features to fulfill requirements. • Adhere to coding best practices. • Overall application functionality, robustness and usability.

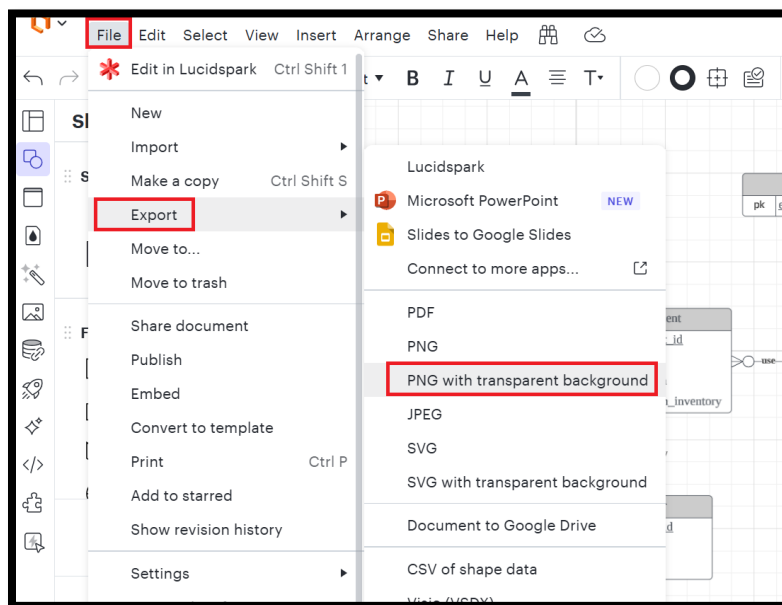
Report and Presentation/Interview 25%	Evaluates the quality of the reporting as well as his/her ability to present and explain the work done. Key Aspects: Report: Clarity, structure, and comprehensiveness of the documentation. Documents necessary evidence for the work done, substantiate with meaningful explanations/discussions of design decisions, solutions and technical considerations. Presentation/Interview: Organization, clarity, and flow of the presentation. Ability to explain key technical aspects. Confident responses to questions, demonstrating deep understanding of concepts.
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-- End of Assignment Specifications --

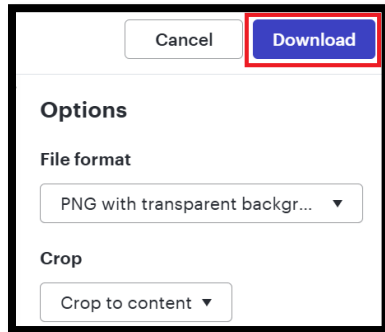
C. Appendix

How to Export Lucid Chart ERD As an Image:

1. Export the file



2. Download the file.



3. Now you can insert the file as image in your report. Ensure it is clear and readable.